LNB GROUP · NAVJYOTI · AKOLA UNIT

Navjyoti VermiTrack

Desktop app deployment & installation process flow

Document	Details
Prepared for	LNB Group IT team
Requested by	Pulkit Singh, DVP (Key Person, VermiTrack)
Purpose	Install the VermiTrack desktop app for all project users and connect it to LNB Group's Microsoft 365 (Entra ID sign-in, SharePoint data, Outlook notifications)
Software	Navjyoti VermiTrack desktop 1.0.0, Windows 10/11 (64-bit) and macOS
Installer	VermiTrack-Windows-Setup.exe (~95 MB) · VermiTrack-Mac.dmg (~200 MB)
Effort for IT	About 30–45 minutes one-time, plus standard software deployment

Contents: 1. What the software does · 2. How it works · 3. End-to-end process flow · Phases 1–6 · 4. Security summary · 5. Troubleshooting · 6. Sign-off checklist

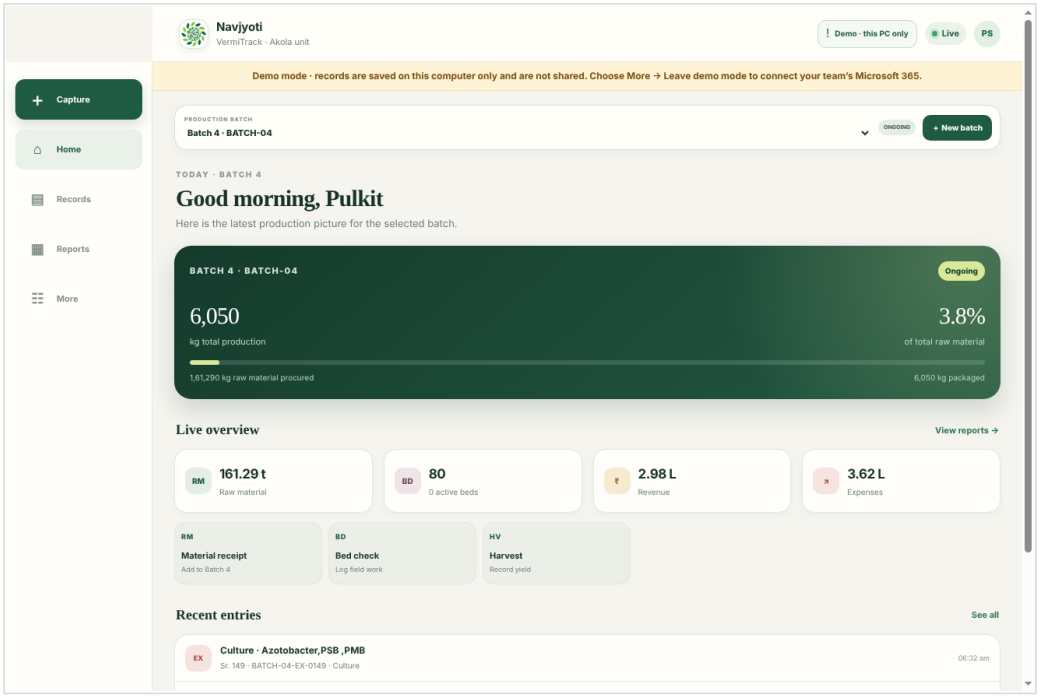
1. What VermiTrack does

VermiTrack replaces the Excel production registers of the Navjyoti vermicompost project with one shared desktop application. Every authorised colleague records data on their own PC, and everyone sees the same live records, reports and charts.

Register	What is captured
Raw material purchase	Supplier, material type, quantity, rate, vehicle/invoice, moisture, temperature, lab sampling. Lot code is generated automatically.
Pre-composting	Lot set-up, culture application, turning, temperature and moisture readings, transfer to beds
Bed production & operations	Bed filling, earthworm application, feeding, watering, status, harvest linkage
Harvesting & packaging	Net yield after sieving, finished batch number, packaging type, packets, moisture
Quality control	Lab report, pH, EC, NPK, organic carbon, micronutrients, FCO compliance (raw material and finished goods)
Sales register	Invoice, customer/dealer, batch, quantity, price, revenue
Stock inventory	Opening stock, production, sales outflow, loss, physical audit
Cost & expenses	Category, vendor, payment mode, cost, optimisation notes

Built-in controls carried over from the existing web version: production batches; batch-wise consolidated reports and charts; CSV export for Excel; automatic serial numbers and register codes; a complete audit trail of every login and change; a 24-hour edit window (after that, only an administrator can edit); deletion only after approval by Pulkit Singh or Pritam Suryavanshi, with a 90-day recycle bin; and per-user limits on registers and batches.

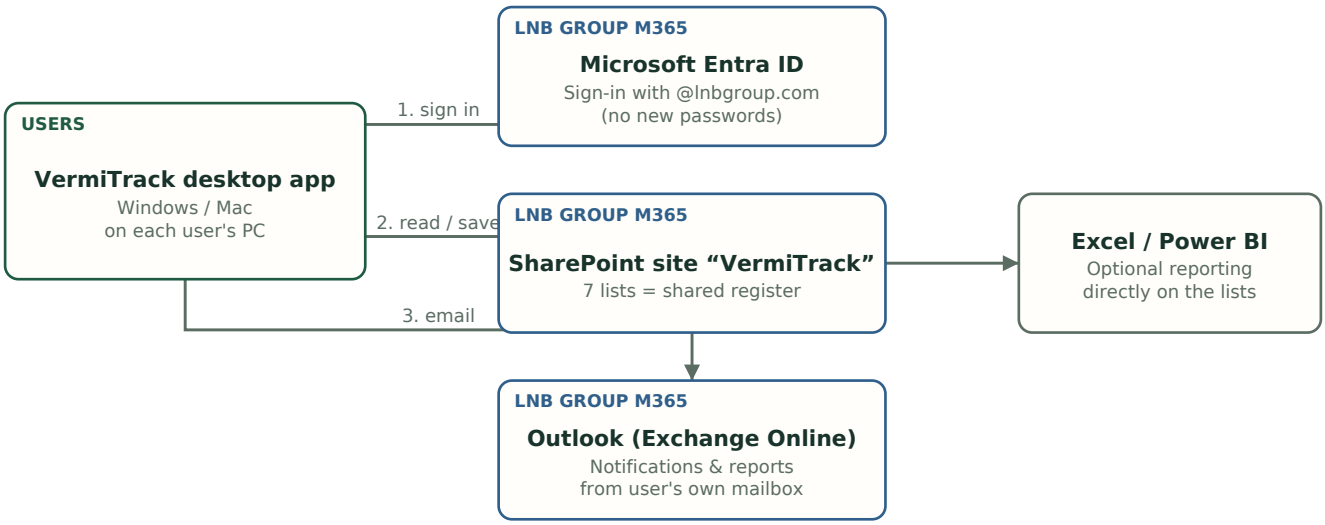
New in the desktop edition: sign-in with the LNB Group Microsoft 365 account; data held on LNB Group's own SharePoint; Outlook email for deletion requests and decisions, user invitations and batch reports.



Home screen: live batch overview with the imported Batch 4 register (screenshot taken in demo mode)

2. How it works

VermiTrack is a desktop application only; **there is no external server or third-party cloud**. It uses three services LNB Group already has in Microsoft 365, through Microsoft Graph:



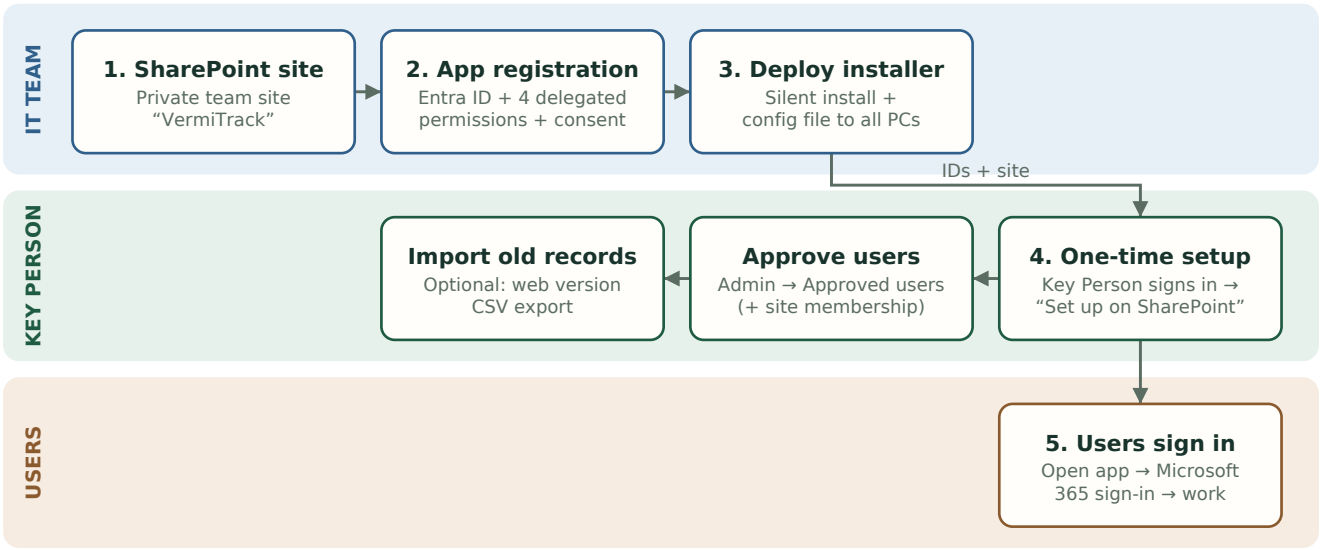
Component	Role	Owner
VermiTrack desktop app	User interface on each PC. Holds no business data, only connection settings and an encrypted sign-in token.	IT deploys
Microsoft Entra ID	Sign-in with existing @lnbgroup.com accounts (MFA and conditional access apply as usual). One app registration.	IT
SharePoint site “VermiTrack”	Stores the register in 7 lists. Site membership controls who can reach the data.	IT creates; Key Person manages
Outlook / Exchange Online	Sends notifications and reports from the signed-in user’s own mailbox (visible in Sent Items).	Existing

SharePoint lists created automatically in Phase 4

List	Contents
VermiTrack Entries	All register entries (8 registers), one item per entry
VermiTrack Batches	Production batches (Batch 3, 4, 5 …)
VermiTrack Baselines	Imported Excel baseline totals
VermiTrack Users	Approved users, roles and register/batch permissions
VermiTrack Activity	Audit trail of every action
VermiTrack Deletion Requests	Deletion requests and approvals
VermiTrack Sign-ins	Sign-in and session log

3. End-to-end process flow

The rollout has six phases. IT completes Phases 1–3 once, the Key Person completes Phases 4 and 6, and each user completes Phase 5 in about two minutes.



#	Activity	Done by	Time	Output
1	Create the private SharePoint team site "VermiTrack" and add members	IT	5 min	Site URL
2	Register the app in Entra ID, add 4 delegated Graph permissions, grant admin consent	IT (Global / Application Admin)	10 min	Client ID + Tenant ID
3	Install the app on users' PCs (manually or silently) with the pre-configured connection file	IT	Standard deployment	App on every PC
4	Sign in, create the lists and load the existing Batch 3–5 data	Key Person	3 min	Live shared register
5	Users open the app and sign in with their Microsoft 365 account	Each user	2 min	User working
6	Approve new users; optionally import records from the old web version	Key Person	As needed	Complete data

PHASE 1 — Create the SharePoint site

IT TEAM
≈ 5 minutes

1. Open **SharePoint admin center** → Sites → Active sites → **+ Create** → **Team site**. Alternatively, use SharePoint home → **+ Create site**.
2. Site name: **VermiTrack**. Privacy: **Private**. Language: English. Owners: **Pulkit Singh** (and IT).
3. Add the project users as **Members** (edit rights). Initial users: Pulkit Singh, Ashok Pawar, Harish Venkatakrishnan, Akola Team, Pritam Suryavanshi. Add view-only users as **Visitors**.
4. Note the site address, for example `https://lnbgroup.sharepoint.com/sites/VermiTrack`.

> Pulkit Singh must be a site **Owner**: creating the lists in Phase 4 needs owner rights. Use a **Private** site; do not grant access to “Everyone except external users”.

PHASE 2 — Register the app in Microsoft Entra ID

IT TEAM
≈ 10 minutes · needs Global / Application Administrator

1. Open **entra.microsoft.com** → Identity → Applications → **App registrations** → **+ New registration**.
2. Name: **Navjyoti VermiTrack**. Supported account types: **Accounts in this organizational directory only (single tenant)**.
3. Redirect URI: platform **Public client/native (mobile & desktop)**, value `http://localhost`. Click **Register**.
4. From the Overview page, copy the **Application (client) ID** and the **Directory (tenant) ID**.
5. Open **API permissions** → + Add a permission → Microsoft Graph → **Delegated permissions**, and add the four permissions in the table below.
6. Click **Grant admin consent for LNB Group** → Yes. All four permissions should show a green “Granted” status.
7. No client secret or certificate is needed (it is a public desktop client).

Permission (delegated)	Purpose	Consent
User.Read	Identify the signed-in user (name, email)	Granted for all users in step 6
Sites.ReadWrite.All	Read and save register items in the VermiTrack lists	Granted for all users in step 6
Sites.Manage.All	Create the 7 lists on the site (first setup only)	Granted for all users in step 6
Mail.Send	Send notifications and reports from the user’s own mailbox	Granted for all users in step 6

> **Delegated permissions only.** The app acts as the signed-in person and can reach only what that person can already reach. A user who is not a member of the VermiTrack site cannot open its lists. There are no application (background) permissions. Optional hardening: under Enterprise applications → Navjyoti VermiTrack → Properties, set Assignment required = Yes and assign the project users or a security group.

PHASE 3	Install the software on users' computers	IT TEAM Standard deployment

3A. Pre-configure the connection (recommended)

Create the file below on each PC (via Intune, GPO or a script). VermiTrack then opens straight to Microsoft 365 sign-in, and users never type IDs or a setup code.

Platform	File location
Windows	`C:\ProgramData\VermiTrack\vermitrack.config.json`
macOS	`/Library/Application Support/VermiTrack/vermitrack.config.json`

```
{
  "clientId": "<Application (client) ID from Phase 2>",
  "tenantId": "<Directory (tenant) ID from Phase 2>",
  "siteUrl": "https://lnbgroup.sharepoint.com/sites/VermiTrack"
}
```

3B. Install the application

Method	How
Silent, all users (recommended)	`VermiTrack-Windows-Setup.exe /S /allusers` (installs to Program Files; Start menu and desktop shortcuts are created)
Silent, current user	`VermiTrack-Windows-Setup.exe /S /currentuser` (no admin rights needed)
Microsoft Intune	Package the .exe as a Win32 app (IntuneWinAppUtil). Install command: `VermiTrack-Windows-Setup.exe /S /allusers`. Detection: file `%ProgramFiles%\Navjyoti VermiTrack\Navjyoti VermiTrack.exe`. Deploy the config file as a separate script or as part of the package.
Manual (single PC)	Double-click the installer → choose for all users or only me → Next → Install → Finish.
macOS	Open VermiTrack-Mac.dmg and drag the app to Applications. On first run, right-click → Open.
Uninstall	Settings → Apps → Navjyoti VermiTrack → Uninstall (silent: run the uninstaller with `/S`).

Code signing: version 1.0 is not yet signed with a code-signing certificate, so Edge and Windows SmartScreen warn that the file is uncommon. In Edge: Downloads → ⋯ → Keep → Show more → Keep anyway. On launch: More info → Run anyway. For organisation-wide rollout, IT may (a) allow the file hash in Defender / SmartScreen, or (b) sign the installer with LNB Group's code-signing certificate. The source code is available for IT review.

Requirement	Detail
Operating system	Windows 10/11 64-bit, or macOS 12+
Disk / memory	≈ 350 MB installed, 4 GB RAM
Network (HTTPS 443)	login.microsoftonline.com, graph.microsoft.com, lnbgroup.sharepoint.com
Local data	%APPDATA%\vermitrack-desktop: settings, encrypted token cache (Windows DPAPI / macOS Keychain) and log. Register data stays in SharePoint.

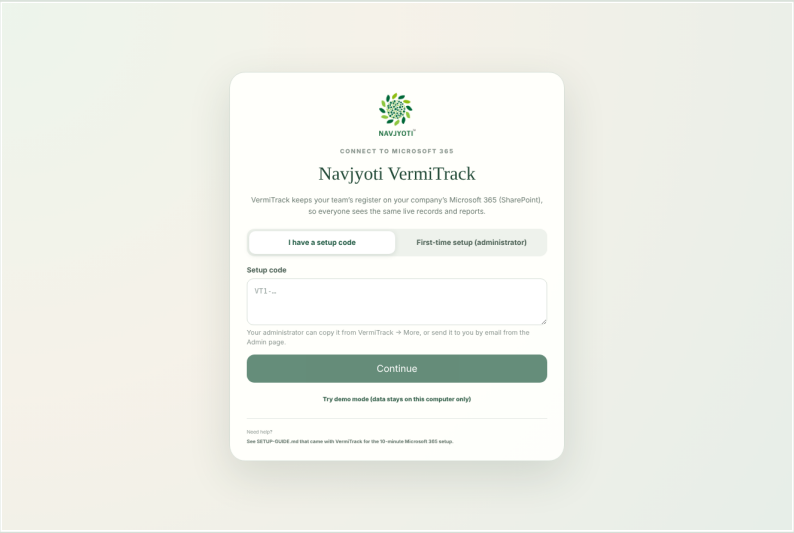
PHASE 4	One-time setup on SharePoint	KEY PERSON (PULKIT SINGH) ≈ 3 minutes · Pulkit Singh

1. Open **Navjyoti VermiTrack** from the Start menu. If IT deployed the config file, it opens on **Sign in with Microsoft 365**. Otherwise choose **First-time setup (administrator)** and enter the Client ID, Tenant ID and site URL.

2. Click **Sign in with Microsoft 365**. The browser opens; sign in as pulkit.singh@lnbgroup.com and return to the app.

3. Click **Set up VermiTrack on this SharePoint site**. The app creates the 7 lists and loads the existing data: 149 Batch 3–5 register entries, the Batch 4 baseline and the 5 approved users.

4. The Home screen appears with the live Batch 4 figures. Setup is complete for the whole organisation.



First-time setup screen (only shown when IT has not pre-configured the connection)

PHASE 5 — User sign-in and daily use

EACH USER
≈ 2 minutes per user

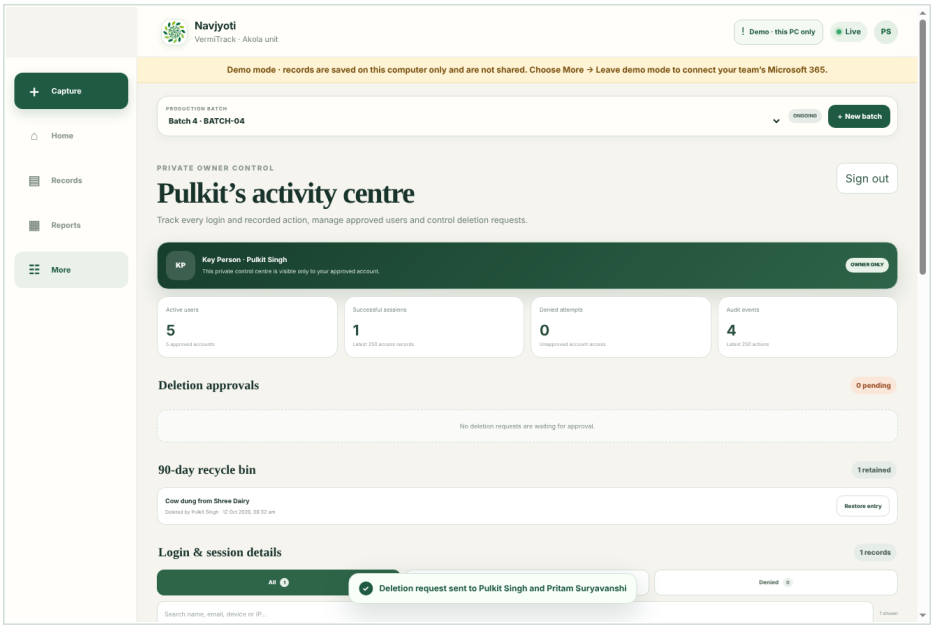
1. The user must be (a) listed in VermiTrack → Admin → **Approved users** and (b) a Member of the VermiTrack SharePoint site.
2. Open VermiTrack → **Sign in with Microsoft 365** → sign in with the @lnbgroup.com account (MFA as usual). The sign-in is remembered.
3. Use **Capture** to record entries, **Records** to search, edit or export, and **Reports** for batch reports, charts, Excel export and ✉ **Email report**. Data refreshes automatically every 20 seconds.

> Without the IT config file, users choose **I have a setup code** and paste the code the Key Person sends from Admin → Approved users → ✉ Email setup instructions.

PHASE 6 — Manage users and bring over existing records

KEY PERSON (PULKIT SINGH)
As needed

1. **Add a user**: Admin (initials, top right) → Approved users → + Add an approved user (email = Microsoft 365 sign-in), choose role, registers and batches. Ask IT to add them to the SharePoint site.
2. **Remove a user**: set the account to Inactive in Approved users, and ask IT to remove them from the site.
3. **Records from the old web version** (optional): in the old web app, Records → All registers → Export CSV for each batch; in the desktop app, Admin → **Bring records from the web version** → choose the file. Duplicates are skipped automatically.



Admin panel: deletion approvals, recycle bin, approved users and audit trail

4. Security summary

Topic	How it is handled
Identity	Microsoft Entra ID sign-in only; LNB Group MFA and conditional access policies apply. No separate passwords.
Data location	LNB Group’s own SharePoint Online tenant. No external servers, databases or third-party cloud services.
Access control	Two layers: SharePoint site membership (IT) and the VermiTrack approved-user list with roles and register/batch limits (Key Person).
Permissions	Delegated Microsoft Graph permissions only; no application permissions or client secrets.
Audit	In-app audit trail of every sign-in, entry, edit, deletion request and approval, plus SharePoint version history on every list item.
Deletion	Two-person rule: deletions need approval by Pulkit Singh or Pritam Suryavanshi; restorable for 90 days.
On the PC	Sign-in token encrypted with Windows DPAPI / macOS Keychain. No register data is kept on the PC.
Point to note	The app enforces roles and approvals. A site Member could still edit list items directly in SharePoint, so keep site membership to the VermiTrack team; such edits show in SharePoint version history.

5. Troubleshooting

Message / symptom	Cause and fix
“Need admin approval” at sign-in	Admin consent not granted. Complete Phase 2, step 6.
“… is not an approved VermiTrack user”	Add the person under Admin → Approved users with their exact sign-in email.
“does not have permission for the VermiTrack SharePoint site”	Add the user as a Member of the SharePoint site.
“Sign-in did not complete”	Check that the redirect URI `http://localhost` is under the **Public client/native** platform; check that the browser can reach login.microsoftonline.com.
List creation fails in Phase 4	The signed-in person must be a site Owner, and Sites.Manage.All must be consented.
SmartScreen / Edge blocks the download	See the code-signing note in Phase 3.
Anything else	File → Open data folder → send vermitrack.log to the Key Person.

6. Sign-off checklist

	IT team
☐	Private SharePoint team site “VermiTrack” created; Pulkit Singh is an Owner
☐	Project users added as Members (view-only users as Visitors)
☐	Entra app registration “Navjyoti VermiTrack” created (single tenant, public client, http://localhost)
☐	4 delegated Graph permissions added and admin consent granted
☐	Client ID, Tenant ID and site URL shared with Pulkit Singh
☐	vermitrack.config.json deployed to target PCs
☐	Installer deployed (/S /allusers) and SmartScreen allowance or code signing handled

	Key Person
☐	Signed in and completed “Set up VermiTrack on this SharePoint site”
☐	Verified Batch 4 figures on the Home and Reports screens
☐	Approved-user list reviewed; roles and permitted registers/batches set
☐	Old web-version records imported (if applicable)
☐	Each user confirmed able to sign in and save a test entry

Role	Name	Signature / date
IT		
Key Person	Pulkit Singh	